

Homework 3

Quadratics and the discriminant · about 60 minutes

Each question graded N0–E8

Show all working. Give answers to 3 decimal places unless the question asks for an exact answer. Each part shows the highest grade it can give: [A], [M] or [E].

Quick review

1 Evaluate $\log_3 81$. [A]

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2 Solve $2^x = 50$. [A]

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3 Simplify $\frac{x^2 - 9}{x + 3}$. [A]

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4 Expand $(x + 5)^2$. [A]

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QUESTION ONE

(a) Solve $x^2 + 2x - 24 = 0$. [A]

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(b) Solve $2x^2 - 3x - 4 = 0$. [A]

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(c) Solve $x(x - 5) = 14$. [A]

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(d) Sam answered two questions. **Sam's answer has 2 errors.** Find each one, fix it, and say why it loses marks. [M]

Sam's answer

(i) Solve $2x^2 + 5x = 3$. $x(2x + 5) = 3$, so $x = 3$ or $2x + 5 = 3$, $x = -1$

(ii) How many real roots has $x^2 - 6x + 9 = 0$? $36 - 36 = 0$, so no real roots

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- (e) Solve $(x^2 - 4x)^2 - 17(x^2 - 4x) + 60 = 0$. [E]

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QUESTION TWO

- (a) Write an equation $x^2 + bx + c = 0$ with roots -3 and 5 . [A]

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- (b) Write an equation with roots $\frac{2}{5}$ and -3 , with whole-number coefficients. [A]

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- (c) A parabola crosses the x -axis at -4 and 1 and passes through $(0, 8)$. Find its equation. [M]

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- (d) Show that the roots of $3x^2 - 10kx + 3k^2 = 0$ ($k \neq 0$) are $3k$ and $\frac{k}{3}$. [M]

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- (e) $x^2 + bx + 18 = 0$ has one root twice the other. Find the possible values of b . [E]

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QUESTION THREE

- (a) Find the discriminant of $3x^2 - 5x + 4 = 0$ and state the number of real roots. [A]

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(b) Find k so that $x^2 - 10x + k = 0$ has equal roots. **[A]**

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(c) Find the values of m for which $2x^2 + mx + 18 = 0$ has no real roots. **[M]**

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(d) $kx^2 + 6x + (k + 8) = 0$ has equal roots. Find k . **[M]**

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(e) Show that $x^2 + (2k - 1)x + k^2 - k = 0$ has two different real roots for every value of k . **[E]**

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